Andrew Paul

University of North Carolina at Chapel Hill
Department of Mathematics
Phillips Hall 315
120 E. Cameron Avenue
Chapel Hill, N.C. 27599 USA

Email: anpa@unc.edu
URL: https://anpaul.onrender.com

Areas of specialization

My training is primarily in geometry and topology, especially complex algebraic geometry. My current study is focused on Hilbert schemes of points, Lagrangian fibrations, and their twists.

Education

2023–Present PH.D. Mathematics; University of North Carolina at Chapel Hill

- Advisor: Justin Sawon

2019–2023 B.S. Mathematics; University of California, San Diego

- Honors with Distinction
- Thesis title: *Chern Classes and Lines on a Cubic Surface*; thesis advisor: Dragos Oprea

Papers

2024 ***Weak convergence of spectral shift functions revisited*** with Carson Connard, Benjamin Ingimarson, and Roger Nichols; *Pure and Applied Functional Analysis* Volume 9(4), 2024, pages 1023–1051. [arXiv: 2211.14970]

2023 ***Topological Properties of Almost Abelian Groups*** with Zhirayr Avetisyan, Oderico-Benjamin Buran, and Lisa Reed. [arXiv: 2308.08059]

Attended Conferences, Workshops, and Summer Schools

2026 International Congress of Mathematicians
2026 ICM Satellite Conference: Conformal Geometry, Einstein Manifolds, and Minimal Submanifolds (Penn State)
2026 Triangle Area Graduate Mathematics Conference (UNC)
2026 Geometry Festival (Duke)
2025 University of Antwerp summer school in Non-Commutative Algebraic Geometry
2024 Palmetto Number Theory Series (Wake Forest)

2024 Graduate Student Topology and Geometry Conference (Michigan State)
2024 Triangle Area Graduate Mathematics Conference (UNC)
2023 Joint Mathematics Meetings
2022 UTC Fall Research Conference
2022 Joint Mathematics Meetings
2021 Gulf Coast Undergraduate Research Symposium (Rice)
2021 Young Mathematicians Conference
2021 Southern California Math REU Conference

Organizational Roles

2026–2027 Co-organizer of the UNC Graduate Mathematics Association Seminar.
2025–Present Organizer of the UNC graduate student algebraic geometry reading group.

Research Experiences

2022 UT Chattanooga Mathematics REU
2021 UC Santa Barbara Mathematics REU
2020–2021 Computational Astrophysics; PI: Thomas Siegert

Teaching

UNC Chapel Hill

Fall 2026 Instructor; MATH 381 (Discrete Mathematics), MATH 231L (Calculus I Corequisite)
Summer 2026 Instructor; MATH 152 (Calculus for Business and Social Sciences)
Spring 2026 Teaching assistant; MATH 231 (Calculus of Functions of One Variable I)
Summer 2025 Instructor; MATH 231 (Calculus of Functions of One Variable I)
Spring 2025 Teaching assistant; MATH 231 (Calculus of Functions of One Variable I), MATH 548 (Combinatorial Mathematics)
Fall 2024 Teaching assistant; MATH 232 (Calculus of Functions of One Variable II), MATH 233 (Calculus of Functions of Several Variables)
Spring 2024 Teaching assistant; MATH 231 (Calculus of Functions of One Variable I), MATH 233 (Calculus of Functions of Several Variables)
Fall 2023 Teaching assistant; MATH 231 (Calculus of Functions of One Variable I)

UC San Diego

Winter 2023 Teaching assistant; MATH 20B (Calculus for Science and Engineering II)
Fall 2022 Teaching assistant; MATH 20A (Calculus for Science and Engineering I)
Winter 2022 Teaching assistant; MATH 20C (Calculus and Analytic Geometry for Science and Engineering)
Fall 2021 Teaching assistant; MATH 20C (Calculus and Analytic Geometry for Science and Engineering)

Directed Reading Program

Fall 2026 Mentoring Matias Relyea in a reading course on algebraic topology, focusing on homology.
Spring 2026 Mentored Luke Davis in a reading course on algebraic topology, with a final presentation on the fundamental group of the genus g orientable surface.
Fall 2025 Mentored Zhouyi Wu in a reading course on combinatorics and representation theory.

Spring 2024 Mentored Yiyang Fan in a reading course on topology and differential geometry, with a final presentation on the Whitney embedding theorem.

Saret & Co. Education
2023–Present Specialized tutor

Chapel Hill Math Circle
2025–Present Advanced-group instructor

San Diego Math Circle
2020–2024 Cauchy-group instructor

Independent Tutoring
2022–Present Tutor

Mathnasium
2018 Tutor

Other academic positions

UNC Chapel Hill
Spring 2026 Grader; MATH 381 (Discrete Mathematics), MATH 522 (Advanced Calculus II)
Spring 2025 Grader; MATH 548 (Combinatorial Mathematics)
Fall 2023 Grader; MATH 381 (Discrete Mathematics)

UC San Diego
Fall 2022 Grader; MATH 150A (Differential Geometry)
Spring 2022 Grader; MATH 20D (Introduction to Differential Equations)

Last updated: September 12, 2026